

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

5. Consider $p = x^3 - 9x^2 + 5x + 4$ and $q = 2x^3 + 4x^2 - 8x + 7$.

- (a) Use p and q to explain how to evaluate the product pq via divide and conquer at the roots of unity.
- (b) Justify the derivation of the recurrence relation for the cost.

/15

Answer:

5. (a) Multiplying two cubic polynomials leads to a polynomial of degree 6, which has 7 coefficients. So we need 8-th roots of unity, denoted by $\omega_0, \omega_1, \dots, \omega_7$.

We write $p(x) = p_{\text{even}}(x) + xp_{\text{odd}}(x) = -9x^2 + 4 + x(x^2 + 5)$,
and $q(x) = q_{\text{even}}(x) + xq_{\text{odd}}(x) = x^2 + 7 + x(2x^2 - 8)$.

The $p_{\text{even}}, p_{\text{odd}}, q_{\text{even}}, q_{\text{odd}}$ are evaluated at the even indexed roots of unity: $\omega_0, \omega_2, \omega_4, \omega_6$, and we add their function values using $x = \omega_i, i = 0, 1, \dots, 7$.

In the last step, we multiply $p(\omega_i)q(\omega_i), i = 0, 1, \dots, 7$, to obtain the value of the product pq at the 8-th roots of unity.

- (b) The cost $T(n)$ to evaluate a polynomial of degree n is reduced to the cost of evaluating 2 polynomials of degree $n/2$. The combination cost is linear in $2n$, which is $O(n)$, so we have the recurrence $T(n) = 2T(n/2) + O(n)$ which leads to $O(n \log(n))$.